

Professional Summary

Data Scientist with a good background in Machine Learning, Financial Time Series Forecasting and Predictive Analytics. Currently completing an MSc in Data Science with research focused on Temporal Fusion Transformers for joint volatility and market direction forecasting on the S&P 500. Interested in developing robust machine learning systems that transform complex financial and business data into actionable insights.

Technical Skills

Programming	Python, SQL
Data Analysis	Pandas, NumPy, Exploratory Data Analysis, Feature Engineering, Statistical Analysis
Machine Learning	Scikit-learn, Regression, Classification, Random Forest, XGBoost, K-Means
Deep Learning	PyTorch, LSTM, Temporal Fusion Transformer, Hugging Face Transformers
Financial ML	Time Series Forecasting, Volatility Forecasting, Market Direction Prediction, Forecast Evaluation
Visualization	Power BI, Tableau, Excel, Matplotlib
Databases	MySQL, Oracle Database
Software Tools	Git, Jupyter Notebook, BeautifulSoup, Docker, Linux

Research Experience

Master Thesis — Bi-Task Financial Forecasting using Temporal Fusion Transformers 2025 – Present *ESGIS*

- Designed two independent Temporal Fusion Transformer architectures for volatility forecasting and market direction classification on S&P 500 financial time series.
- Built an end-to-end forecasting framework covering financial data acquisition, feature engineering, statistical validation and probabilistic deep learning.
- Engineered more than sixty financial, technical and macroeconomic variables from market and economic data.
- Benchmarked Transformer-based forecasting models against EWMA, ARIMA, LSTM and XGBoost baselines.
- Implemented the project using Python, PyTorch Forecasting, Scikit-learn and Optuna.

Professional Experience

Independent Mathematics Teacher

Sep 2023 – Present

Lomé, Togo

- Teach mathematics, probability and statistics following the competency-based curriculum for secondary education.
- Strengthen analytical reasoning and quantitative problem-solving skills through real-world statistical applications.
- Prepare students for national examinations while developing structured mathematical thinking.

Pharmacy Manager

Sep 2019 – Jul 2023

Kara, Togo

- Managed daily pharmacy operations including inventory, procurement and sales monitoring.
- Performed exploratory analysis on sales and inventory data to improve stock management and reduce shortages.
- Designed Excel dashboards to monitor KPIs and automate weekly operational reporting.
- Maintained MySQL databases supporting inventory management and operational decision-making.
- Implemented cost-control procedures contributing to improved operational efficiency.

Education

Master of Science in Data Engineering

2024 – 2026

ESGIS, Lomé - TOGO

- Research specialization in Machine Learning, Financial Time Series Forecasting and Deep Learning.
- Master Thesis: Bi-Task Financial Forecasting using Specialized Temporal Fusion Transformers.

Bachelor of Science in Mathematics

2016 – 2022

University of Kara

Kara, Togo

- Strong foundation in linear algebra, probability, statistics, optimization and mathematical modelling.

Selected Projects

Pushed Github Project 1	github Link
Pushed Github Project 2	github Link
Pushed Github Project 3	github Link

Research Interests

Time Series Forecasting
Financial Machine Learning
Financial Deep Learning
Natural Language Processing
Applied AI

Professional Certifications

Python for Data Science	Self-paced <i>Codecademy Professional Certificates</i>
Data Analytics	Self-paced <i>Codecademy Professional Certificates</i>
Tableau for Data Visualization	Self-paced <i>Codecademy Professional Certificates</i>
Web Scraping with BeautifulSoup	Self-paced <i>Codecademy Professional Certificates</i>

Languages

French	Native
English	Intermediate
Ewe	Fluent
Kabye	Native